Checkpoint B | Version 5 | Solutions

*Each skill earns **Pass** or **Not Yet**. Each question here assesses one skill. Pass (high) and Pass (low) mark the range of passing work; both earn the same Pass.*

Selling out to Big Chocolate

Many in the wizarding world enjoy chocolate bars both as a snack and as an ingredient. Chocolate is easy to produce, and many small chocolate shops have emerged to make and sell the good. The supply and demand curves for chocolate bars can be represented by the following equations:

$$D: P = 10 - \frac{1}{2}Q_d \qquad S: P = \frac{1}{2}Q_s$$

Price is measured in Coins and quantity is measured in Kilograms.

B1.1 | Demand

Graph the demand curve and calculate Quantity Demanded (Q_d) and Marginal Benefit (MB) at:

What to look for:

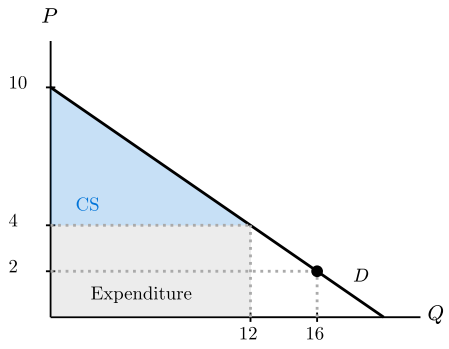
B1.1 Pass (high): Demand drawn sloping down from the right vertical intercept, with Qd read at the price and MB read at the quantity.

B1.1 Pass (low): One arithmetic slip, or a missing axis label, when the graph and the other answer show the skill.

B1.1 Not Yet: Demand sloping up, Qd and MB swapped (reading a quantity as a price), or no graph.

a) Q_d at 4 Coins: **12**

b) MB of the 16th unit: **2**



Demand: $10 - 0.5Q$, vertical intercept 10, horizontal intercept 20.

a) $4 = 10 - 0.5Q$, so Qd = **12**.

b) MB $= 10 - 0.5 \cdot 16 =$ **2**.

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B1.2 | Consumer Surplus

Shade Consumer Surplus (*CS*) and Expenditure at 4 Coins.

What to look for:

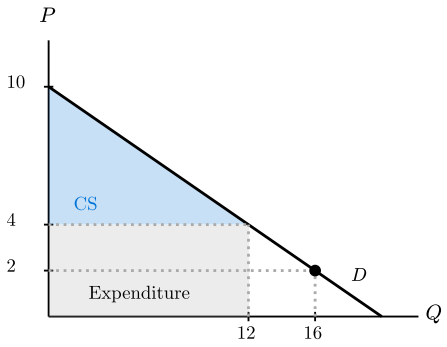
B1.2 Pass (high): CS shaded as the triangle below demand and above the price, expenditure as the rectangle below the price, both computed.

B1.2 Pass (low): One arithmetic slip with the regions right.

B1.2 Not Yet: CS and expenditure swapped, CS measured below the price, or no areas.

a) *CS*: **36**

b) *Expenditure*: **48**



Demand: $10 - 0.5Q$, vertical intercept 10, horizontal intercept 20.

a) CS = ½ · 12 · (10 − 4) = **36**.

b) Expenditure = 4 · 12 = **48**, the rectangle under the price.

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B2.1 | Supply

Graph the supply curve and calculate Quantity Supplied (Q_s) and Marginal Cost (MC) at:

What to look for:

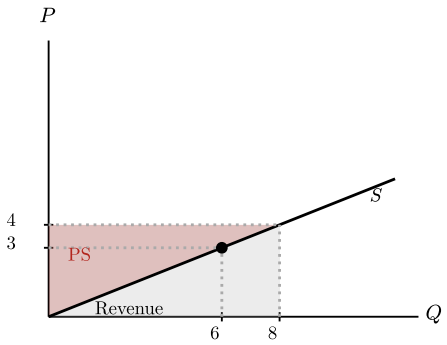
B2.1 Pass (high): Supply drawn sloping up from its intercept, with Qs read at the price and MC read at the quantity.

B2.1 Pass (low): One arithmetic slip, or a missing axis label, when the graph and the other answer show the skill.

B2.1 Not Yet: Supply sloping down, Qs and MC swapped, or no graph.

a) Q_s at 4 Coins: **8**

b) MC of the 6th unit: **3**



Supply: 0.5Q, vertical intercept 0.

a) $4 = 0.5Q$, so Qs = **8**.

b) MC $= 0.5 \cdot 6 =$ **3**.

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B2.2 | Producer Surplus

Shade Producer Surplus (*PS*) and Revenue at 4 Coins.

What to look for:

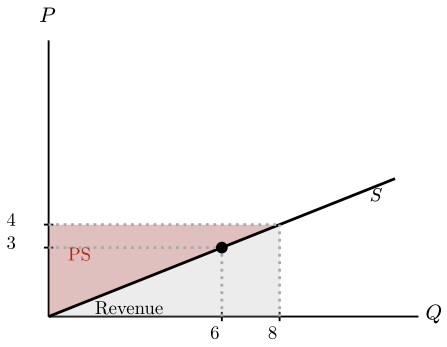
B2.2 Pass (high): PS shaded as the triangle above supply and below the price, revenue as the rectangle below the price, both computed.

B2.2 Pass (low): One arithmetic slip with the regions right.

B2.2 Not Yet: PS and revenue swapped, PS measured above the price, or no areas.

a) *PS*: **16**

b) *Revenue*: **32**



Supply: 0.5Q, vertical intercept 0.

a) PS = ½ · 8 · (4 − 0) = **16**.

b) Revenue = 4 · 8 = **32**.

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B3.1 | Equilibrium

Use the graph and algebra to find equilibrium price and quantity.

What to look for:

B3.1 Pass (high): Equilibrium from setting quantity demanded equal to quantity supplied, the shortage or excess named and sized, and the price moving in the direction that closes it.

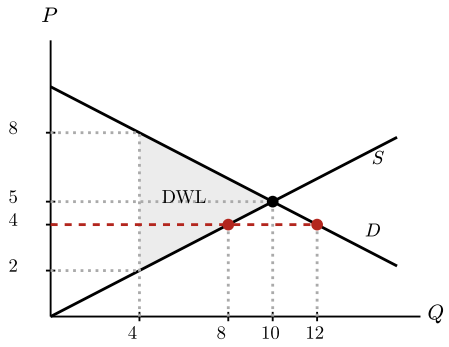
B3.1 Pass (low): An arithmetic slip carried through consistently.

B3.1 Not Yet: Shortage and excess swapped, or a price direction that would widen the gap.

a) Equilibrium Price: **5** Equilibrium Quantity: **10**

b) At 4 Coins: **shortage**, excess (circle one) of size **4**

c) From 4 Coins, the price will: **rise**, fall



a) $10 - 0.5Q = 0.5Q$ gives $Q^* =$ **10** and $P^* =$ **5**.

b) At 4, $Q_d = 12$ and $Q_s = 8$: a **shortage** of **4**.

c) The price will **rise**: buyers who miss out bid it up.

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B4.1 | Efficiency

Use a graph to describe the market for chocolate and its associated welfare.

What to look for:

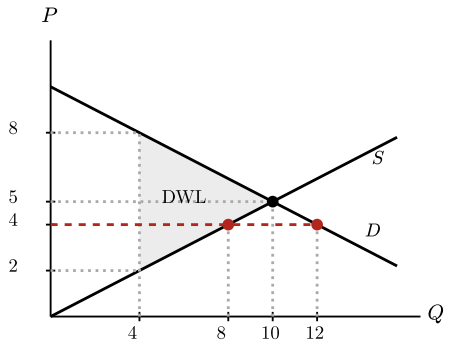
B4.1 Pass (high): MB and MC read at the restricted quantity, MB above MC recognized as room for more surplus, and the deadweight loss as the triangle between the curves from that quantity to equilibrium.

B4.1 Pass (low): One arithmetic slip with the region right, or MB and MC right with the triangle set up but not finished.

B4.1 Not Yet: A deadweight loss on the wrong side of equilibrium, or the whole area between the curves taken as the loss.

a) At 4 Kilograms, MB: **8** MC: **2**

b) Deadweight loss if only 4 are exchanged: **18**



a) At Q = 4, MB = 10 − 2 = **8** and MC = 2 = **2**. MB exceeds MC, so every unit from the 5th to the 10th would add surplus.

b) DWL = ½ · (8 − 2) · (10 − 4) = **18**, the triangle between the curves from 4 to 10.

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B4.2 | Price Controls

Use a graph and algebra to describe the market for chocolate and the welfare effects if a price floor of 7 Coins was imposed on this vibrant market for chocolate.

What to look for:

B4.2 Pass (high): The floor recognized as binding, the quantity exchanged as the smaller of Qd and Qs, the excess sized, CS placed and computed as the triangle above the floor out to the quantity traded, and buyers seen to lose.

B4.2 Pass (low): One quantity slip carried through consistently, or the CS triangle right with an arithmetic slip.

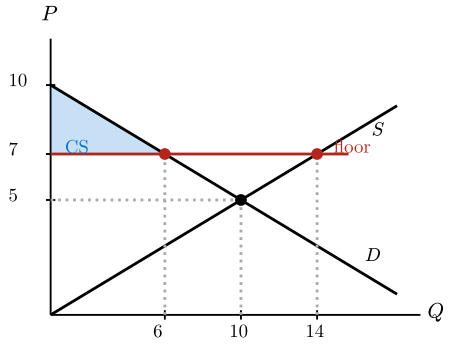
B4.2 Not Yet: The larger quantity used as the quantity exchanged, the floor treated as not binding, or CS measured below the floor.

a) Quantity Demanded: **6** Supplied: **14** Exchanged: **6**

b) **Excess**, Shortage (circle one) of size **8**

c) Consumer Surplus (*CS*): **9**

d) Compared with equilibrium, *CS*: increases, **shrinks** (circle one)



The floor of 7 is above P* = 5, so it binds.

Qd: $7 = 10 - 0.5Q$ gives **6**. Qs: $7 = 0.5Q$ gives **14**. Exchanged: the smaller, **6**. **Excess** of **8**.

CS = ½ · 6 · (10 − 7) = **9**, the triangle above the floor out to the 6 units traded.

Buyers **lose**: they pay more (7 instead of 5) for fewer units (6 instead of 10). No computation is needed for the direction.

{++*CS* **shrinks**, from 25 at equilibrium to 9 under the floor.++}

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B5.1 | Shifters

Draw and label on the graph how the curves shifted after two separate events: ***First***, a new agricultural technology improves the harvesting efficiency of cocoa, a component of chocolate bars. ***Second***, doctors announce public concern over the relationships between sugar content in chocolate bars and metabolic disorders.

What to look for:

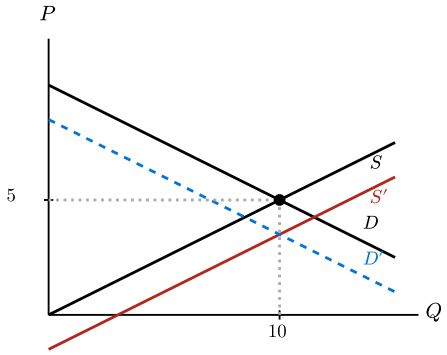
B5.1 Pass (high): The right curve and direction for both events, with the shifted curves drawn. Up and down are read on the price axis: supply down is supply out, demand down is demand in. In, out, left, and right are accepted when they match.

B5.1 Pass (low): One direction mislabeled when the graph shows the right shift.

B5.1 Not Yet: A movement along a curve treated as a shift, or the wrong curve.

a) After the new tech, which curve shifted? **Supply** Up, **Down** (circle one)

b) After the announcement, which curve? **Demand** Up, **Down** (circle one)



a) **Supply**, **Down**: better harvesting lowers the cost of cocoa, so sellers offer each quantity at a lower price (supply shifts down, or out).

b) **Demand**, **Down**: the health warning makes chocolate less attractive, so buyers want less at every price (demand shifts down, or in).

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B5.3 | Comparative Statics

After both events in B5.1, circle what happens to the equilibrium price and quantity.

What to look for:

B5.3 Pass (high): For the two shifts together, the one change that is determined and the one that is indeterminate, read from the directions of the shifts.

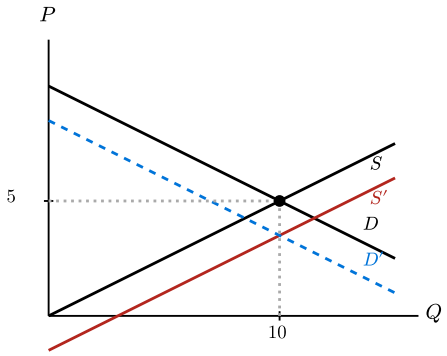
B5.3 Pass (low): One of the two reported the wrong way when the graph shows both shifts.

B5.3 Not Yet: Both changes reported as determined, or the determined change in the wrong direction.

Grading B5.3: Grade B5.3 against the shifts the student gave in B5.1. If those shifts are wrong, B5.3 passes when its price and quantity answers follow correctly from them.

a) Price: increased, stayed the same, **decreased**, indeterminate (circle one)

b) Quantity: increased, stayed the same, decreased, **indeterminate** (circle one)



From B5.1, supply shifts down and demand shifts down. Price **decreased**, quantity **indeterminate**: both shifts push the price the same way, while their effects on quantity run opposite, so the net change in quantity depends on the sizes of the shifts.

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B5.2 | Elasticity

Use the midpoint method with numbers in the table, then circle the classification.

What to look for:

B5.2 Pass (high): Both midpoint elasticities with the right classifications, demand classified by its absolute value.

B5.2 Pass (low): One arithmetic slip with the method and classifications consistent with the numbers.

B5.2 Not Yet: Slope used in place of elasticity, a classification that contradicts the number, or no midpoint averages.

a) Price Elasticity of Demand between 4 and 8 coins: **−1.5 elastic**, unit elastic, inelastic

b) Price Elasticity of Supply between 4 and 8 coins: **1** elastic, **unit elastic**, inelastic

P	Qd	Qs
4	12	8
8	4	16

a) Qd goes from 12 to 4. $(-8/8) \div (4/6) = \mathbf{-1.5}$, **elastic** ($|-1.5| > 1$).

b) Qs goes from 8 to 16. $(8/12) \div (4/6) = \mathbf{1}$, **unit elastic**. A straight-line supply curve through the origin is unit elastic everywhere: quantity and price move in the same proportion.